Final Exam (Spring 2024)

ELEC70082 – Distributed Optimisation and Learning

Instructor: Stefan Vlaski

09/05/2024, 10:00 – 13:00

Read all of the following information before starting the exam:

- The exam is open book. Feel free to use any written or printed material you have brought. You can use and refer to any results from the lecture notes that might be useful in answering the questions.
- Calculators or other electronic devices are not allowed for this exam.
- There is a total of 3 problems that total to 100 marks. There is a total of 4 pages.
- You have 3 hours (10:00 - 13:00) to complete the exam.

Question 1. The following are true/false questions. Full credit will only be given for correct answers with correct justification, but partial credit is possible.

(a) [8 marks] (true/false) The consensus + innovations algorithm is bias-free when agents are homogenous, i.e., when $w_k^o = w^o$ for all k.

Solution. [**4 marks**] The statement is true.

[**4 marks**] The bias of penalty-based algorithms was studied in the lectures, and shown to be proportional to $\nabla_w J_k(w^o)$. This quantity is zero when agents are homogenous.

(b) [8 marks] (true/false) Suppose you are provided with a doubly-stochastic combination matrix $A \in \mathbb{R}^{10\times 10}$, and find its set of eigenvalues:

$$\{1, 1, 0.9, 0.9, 0.5, 0.5, 0.4, 0.3, 0.2, 0.1\} \tag{1}$$

From this, you can conclude that the graph is strongly-connected.

Solution. [**4 marks**] The statement is false.

[**4 marks**] The second largest eigenvalue of a combination matrix of a strongly-connected graph is strictly less than one in magnitude.

(c) [8 marks] (true/false) In the presence of gradient noise, the difference in performance between penalty-based and primal-dual algorithms becomes negligible for very small step-sizes.

Solution. [**4 marks**] The statement is true.

[**4 marks**] Comparing the performance expressions of penalty-based and primal-dual algorithms, the difference appears in higher-order terms in the step-size. This means that for small step-sizes, the difference becomes negligible. See also the discussion at the beginning of Chapter 10.

(d) [8 marks] (true/false) The implementation of a gradient-tracking based algorithm for decentralized optimisation requires a larger number of communication exchanges than a primal-dual or penalty-based algortihm.

Solution. [**4 marks**] The statement is true.

[**4 marks**] Gradient-tracking algorithms rely addtionally on the propagation of the gradient-tracking variables $\boldsymbol{g}_{k,i}$, which roughly doubles the communication cost.

(e) [8 marks] (true/false) Consider 2 different learning setups. In setup one, a single agent is running a stochastic gradient algorithm using mini-batch gradient approximations with $B = 100$. In setup two, $K = 50$ agents are running a consensus+innovations algorithm using mini-batch gradient approximations with $B = 2$. Otherwise, all agents, problem and learning parameters are identical, data is independently and identically distributed, and the step-size is small enough so that higher-order terms can be neglected. Then, the performance of the two setups in terms of their steady-state error will be approximately the same.

Solution. [**4 marks**] The statement is true.

[**4 marks**] Computing the gradient noise constants shows that in both cases the noise of the ordinary gradient approximation is reduced by a factor of 100.

Question 2. We consider a collection of K agents, where each agent k is equipped with a local logistic risk:

$$J_k(w) \triangleq \mathbb{E} \ln \left(1 + e^{-\boldsymbol{\gamma}_{k,i} \boldsymbol{h}_{k,i}^{\mathsf{T}} w}\right) + \frac{\rho}{2}\|w\|^2 \tag{2}$$

We are interested in solving the consensus optimization problem:

$$w^o \triangleq \arg \min_{w \in \mathbb{R}^M} \frac{1}{K} \sum_{k=1}^{K} J_k(w) \tag{3}$$

(a) [10 marks] Suppose each agent sees B observations $\{\boldsymbol{h}_{k,i,b}, \boldsymbol{\gamma}_{k,i,b}\}_{b=1}^{B}$ at time i. All observations (across space and time) are independent. Formulate the recursions for the diffusion and for the exact diffusion algorithm, when agents are employing local mini-batch approximations of the stochastic gradient, each with size B.

Solution. [5 marks] The diffusion recursions for logistic regression with mini-batch stochastic gradients amount to:

$$\boldsymbol{\psi}_{k,i} = (1 - \mu\rho)\boldsymbol{w}_{k,i-1} + \frac{\mu}{B} \sum_{b=1}^{B} \frac{\boldsymbol{\gamma}_{k,i,b} \boldsymbol{h}_{k,i,b}}{1 + e^{\boldsymbol{\gamma}_{k,i,b} \boldsymbol{h}_{k,i,b}^{\mathsf{T}} \boldsymbol{w}_{k,i-1}}} \tag{4}$$

$$\boldsymbol{w}_{k,i} = \sum_{\ell \in \mathcal{N}_k} a_{\ell k} \boldsymbol{\psi}_{\ell,i} \tag{5}$$

[5 marks] Exact diffusion additionally involves a correction step:

$$\boldsymbol{\psi}_{k,i} = (1 - \mu\rho)\boldsymbol{w}_{k,i-1} + \frac{\mu}{B} \sum_{b=1}^{B} \frac{\boldsymbol{\gamma}_{k,i,b} \boldsymbol{h}_{k,i,b}}{1 + e^{\boldsymbol{\gamma}_{k,i,b} \boldsymbol{h}_{k,i,b}^{\mathsf{T}} \boldsymbol{w}_{k,i-1}}} \tag{6}$$

$$\boldsymbol{\phi}_{k,i} = \boldsymbol{\psi}_{k,i} + \boldsymbol{w}_{k,i-1} - \boldsymbol{\psi}_{k,i-1} \tag{7}$$

$$\boldsymbol{w}_{k,i} = \sum_{\ell \in \mathcal{N}_k} a_{\ell k} \boldsymbol{\phi}_{\ell,i} \tag{8}$$

(b) [10 marks] Determine a bound for the limiting performance $\limsup_{i\to\infty} \mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2$ in terms of algorithm parameters, strong-convexity and Lipschitz constants of $J(w)$, locally optimal models w_k^o, mixing rate λ_2 and data statistics for both the diffusion and the exact diffusion algorithm.

Solution. [5 marks] The performance expressions follow from Theorems 9.1 and 10.1, and the discussion at the beginning of Chapter 10. The diffusion algorithm satisfies (Theorem 9.1):

$$\limsup_{i\to\infty} \mathbb{E}\|\widetilde{\boldsymbol{w}}_i\|^2 = O\left(\frac{\mu\sigma^2}{\nu} + \frac{\mu^2\lambda_2^2 \sum_{k=1}^{K} \|\nabla J_k(w^o)\|^2}{(1-\lambda_2)^2} + \frac{\mu^2\lambda_2^2 K^2 \sigma^2}{1-\lambda_2}\right) \tag{9}$$

On the other hand, the exact diffusion algorithms satisfies (Theorem 10.1):

$$\limsup_{i\to\infty} \mathbb{E}\|\widetilde{\boldsymbol{w}}_i\|^2 = O\left(\frac{\mu\sigma^2}{\nu} + \frac{\mu^2\lambda_2^2 K^2 \sigma^2}{1-\lambda_2}\right) \tag{10}$$

[5 marks] In both cases, the missing quantity is σ^2. We find:

$$\sigma^2 = \frac{1}{B} \sum_{k=1}^{K} \sigma_k^2 = \frac{\mathrm{Tr}(R_{\boldsymbol{h}_k})}{B} \tag{11}$$

Pluggin this relation into both performance bounds gives the result.

(c) [5 marks] Simplify the expressions to the case where agents are homogenous and $w_k^o = w^o$. Compare the performance and comment.

Solution. [5 marks] In this case $\|\nabla J_k(w^o)\|^2 = 0$ and the expressions coincide. This is to be expected, since bias-correction, as it appears in exact diffusion, only really makes a difference in heterogeneous environments.

Question 3. We examine a variant of the consensus averaging algorithm over noisy links:

$$\boldsymbol{w}_{k,i} = \sum_{\ell \in \mathcal{N}_k} a_{\ell k} \boldsymbol{w}_{\ell,i-1} + \boldsymbol{v}_{k,i} \tag{12}$$

Each agent is initialized at some deterministic local signal g_k, i.e., $w_{k,0} = g_k$. The combination weights are collected in a matix A that satisfies:

$$A = A^{\mathsf{T}} \tag{13}$$

$$A\mathbb{1} = \mathbb{1} \tag{14}$$

The communication noise $\boldsymbol{v}_{k,i}$ is modelled as zero-mean with $\mathbb{E}\|\boldsymbol{v}_{k,i}\|^2 = \sigma_{v_k}^2$. You may assume $\boldsymbol{v}_{k,i}$ to be independent across agents, and the iterates $\boldsymbol{w}_{k,i-1}$.

(a) [10 marks] Determine a recursion for the evolution of the network centroid:

$$\boldsymbol{w}_{c,i} \triangleq \frac{1}{K} \sum_{k=1}^{K} \boldsymbol{w}_{k,i} \tag{15}$$

Solution. [10 marks] Since A is doubly-stochastic, it holds that:

$$\begin{aligned} \boldsymbol{w}_{c,i} &= \frac{1}{K} \sum_{k=1}^{K} \boldsymbol{w}_{k,i} = \frac{1}{K} \sum_{k=1}^{K} \sum_{\ell \in \mathcal{N}_k} a_{\ell k} \left(\boldsymbol{w}_{\ell,i-1} + \boldsymbol{v}_{\ell,i} \right) \\ &= \frac{1}{K} \sum_{\ell=1}^{K} \sum_{k=1}^{K} \boldsymbol{a}_{\ell k,i} \left(\boldsymbol{w}_{\ell,i-1} + \boldsymbol{v}_{k,i} \right) = \boldsymbol{w}_{c,i-1} + \frac{1}{K} \sum_{k=1}^{K} \boldsymbol{v}_{k,i} \end{aligned} \tag{16}$$

(b) [15 marks] Determine a bound for the expected squared deviation of the local estimates $\boldsymbol{w}_{k,i}$ from the centroid $\boldsymbol{w}_{c,i}$ as $i \to \infty$, captured in the quantity:

$$\limsup_{i \to \infty} \sum_{k=1}^{K} \mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{w}_{c,i}\|^2 \tag{17}$$

Solution. [5 marks] Since A is doubly-stochastic, it holds that:

$$\sum_{k=1}^{K} \|\boldsymbol{w}_{k,i} - \boldsymbol{w}_{c,i}\|^2 = \left\| \left(I - \frac{1}{K} \mathbb{1}\mathbb{1}^{\mathsf{T}} \otimes I \right) \boldsymbol{w}_i \right\|^2 \tag{18}$$

and hence

$$\begin{aligned} \left(I - \frac{1}{K} \mathbb{1}\mathbb{1}^{\mathsf{T}} \otimes I \right) \boldsymbol{w}_i &= \left(I - \frac{1}{K} \mathbb{1}\mathbb{1}^{\mathsf{T}} \otimes I \right) \mathcal{A}^{\mathsf{T}} \left(\boldsymbol{w}_{i-1} + \boldsymbol{v}_i \right) \\ &= \left(\mathcal{A}^{\mathsf{T}} - \frac{1}{K} \mathbb{1}\mathbb{1}^{\mathsf{T}} \otimes I \right) \left(I - \frac{1}{K} \mathbb{1}\mathbb{1}^{\mathsf{T}} \right) \boldsymbol{w}_{i-1} + \left(\mathcal{A}^{\mathsf{T}} - \frac{1}{K} \mathbb{1}\mathbb{1}^{\mathsf{T}} \otimes I \right) \boldsymbol{v}_i \end{aligned} \tag{19}$$

[5 marks] Squaring, taking expectations and removing cross-terms:

$$\sum_{k=1}^{K} \mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{w}_{c,i}\|^2 \leq \lambda_2^2 \sum_{k=1}^{K} \mathbb{E}\|\boldsymbol{w}_{k,i-1} - \boldsymbol{w}_{c,i-1}\|^2 + \lambda_2^2 \sum_{k=1}^{K} \sigma_{v_k}^2 \tag{20}$$

[5 marks] Iterating, we can find:

$$\limsup_{i\to\infty} \sum_{k=1}^{K} \mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{w}_{c,i}\|^2 \leq \frac{\lambda_2^2}{1-\lambda_2^2} \sum_{k=1}^{K} \sigma_{v_k}^2 \tag{21}$$

(c) [10 marks] Determine an expression for the expected squared deviation of the centroid from average, i.e., for:

$$\mathbb{E}\left\| \boldsymbol{w}_{c,i} - \frac{1}{K}\sum_{k=1}^{K} g_k \right\|^2 \tag{22}$$

Based on your conclusion, what can you say about the robustness of the average consensus algorithm to noisy links?

Solution. [5 marks] Using the result from (a), we find:

$$\begin{aligned}
\mathbb{E}\|\boldsymbol{w}_{c,i} - \overline{g}\|^2 &= \mathbb{E}\|\boldsymbol{w}_{c,i-1} - \overline{g} + \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{v}_{k,i}\|^2 \\
&= \mathbb{E}\|\boldsymbol{w}_{c,i-1} - \overline{g}\|^2 + \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}\|\boldsymbol{v}_{k,i}\|^2
\end{aligned} \tag{23}$$

[5 marks] Upon iterating:

$$\mathbb{E}\|\boldsymbol{w}_{c,i} - \overline{g}\|^2 = \|w_{c,0} - \overline{g}\|^2 + \frac{i}{K}\sum_{k=1}^{K} \sigma_{v_k}^2 = \frac{i}{K}\sum_{k=1}^{K} \sigma_{v_k}^2 \tag{24}$$

We observe that the error of the centroid grows with i. As such, the average consensus algorithm is not robust to link noise.